Exercises on the Inclusion–Exclusion Principle

Corrected and expanded problem set · 22 exercises

Conventions. Write $[n] = \{1, \dots, n\}$. All sets are finite, and all people are distinguishable. Empty sums are 0 and empty products are 1; use $0! = 1$ and $0^0 = 1$ in counting formulas. In sums of binomial coefficients, set $\binom{a}{b} = 0$ unless $0 \le b \le a$ for integers a, b. The labels H, HH, and HHH indicate increasing levels of challenge.

1. **Euler's totient function.** For a positive integer n, let $\varphi(n) = |\{k \in [n] : \gcd(k, n) = 1\}|$. If $p_1, \dots, p_m$ are the distinct prime divisors of n, prove that

$$\varphi(n) = n \prod_{i=1}^{m} \left(1 - \frac{1}{p_i}\right).$$

   Include the case $n = 1$.

2. **Surjections and Stirling numbers.** Let $m \ge n \ge 1$. Count the surjections from $[m]$ to $[n]$. If $S(m, n)$ is the number of partitions of $[m]$ into n nonempty, unlabeled blocks, explain why the answer is $n!S(m, n)$ and hence obtain an alternating-sum formula for $S(m, n)$.

3. **Derangements.** A derangement of $[n]$ is a permutation σ with $\sigma(i) \ne i$ for every i. For $n \ge 1$, find an inclusion–exclusion formula for the number D_n of derangements. Set $D_0 = 1$.

4. **Bounded compositions.** Let $n, s \ge 1$ and $k \ge 0$. The number of nonnegative integer solutions of $x_1 + \cdots + x_n = k$ is $\binom{n+k-1}{n-1}$. How many satisfy $x_i < s$ for every $i \in [n]$? Give an alternating-sum formula valid also when $k > n(s-1)$.

5. **Exactly k fixed points.** For $n \ge 1$ and $0 \le k \le n$, count the permutations of $[n]$ with exactly k fixed points. Express the answer using the derangement numbers from Problem 3. In particular, how many have exactly one fixed point?

6. **Dance partners.** There are $n \ge 1$ married couples, each consisting of one man and one woman. Count the ways to divide the $2n$ people into n unordered pairs if every pair contains one man and one woman and no spouses are paired together. How does the answer change if any two people may be paired, still with no spouses together? The collection of pairs is unordered in both cases.

7. **Couples in a row** (H). Seat $n \ge 1$ couples in a row of $2n$ numbered seats. Count the seatings in which no spouses occupy consecutive seats. *Hint.* For a chosen set of couples required to sit together, treat each couple as a block and account for its two internal orders.

8. **Alternating seats in a row** (HH). In Problem 7, suppose each couple consists of one man and one woman, and require men and women to alternate. Count the valid seatings, allowing either a man or a woman in the first seat.

9. **The ménage problem** (HHH). For $n \ge 3$, seat n couples around a circular table so that men and women alternate and no spouses sit next to each other, including across the last and first positions. Count arrangements up to rotation; reflected arrangements are considered distinct. Each couple consists of one man and one woman.

10. Partitions and equivalence relations. For $n \geq 1$ and $1 \leq k \leq n$, count the partitions of $[n]$ into k nonempty, unlabeled groups, using Problem 2. Explain why this also counts equivalence relations on $[n]$ with exactly k classes. Express the total number of equivalence relations on $[n]$ as a sum of Stirling numbers.

11. Cancellation on an interval of subsets. For finite sets $I \subseteq J$, prove that

$$\sum_{I \subseteq K \subseteq J} (-1)^{|K \setminus I|} = \begin{cases} 1, & I = J, \\ 0, & I \neq J. \end{cases}$$

Give both a binomial-theorem proof and a proof by pairing cancelling terms.

12. An inverse Pascal matrix. Let $n \geq 0$. Define $(n+1) \times (n+1)$ matrices, with rows and columns indexed by $0, 1, \ldots, n$, by

$$A_{ij} = \binom{i}{j}, \qquad B_{ij} = (-1)^{i+j} \binom{i}{j}.$$

Use inclusion–exclusion, or the cancellation in Problem 11, to show that $AB = I_{n+1}$.

13. A bijective proof of inclusion–exclusion. Let $A_1, \ldots, A_n \subseteq U$, where U is finite and $n \geq 1$. For $I \subseteq [n]$, set $A_I = \bigcap_{i \in I} A_i$, with $A_\varnothing = U$. Prove bijectively that

$$\left| U \setminus \bigcup_{i=1}^{n} A_i \right| = \sum_{I \subseteq [n]} (-1)^{|I|} |A_I|.$$

Move all negative terms to the left and realize both sides as cardinalities of disjoint unions. Give a bijection between these unions. *Hint.* For $x \in \bigcup_i A_i$, toggle the least index i for which $x \in A_i$.

14. Perfect matchings and the permanent. Let $n \geq 1$ and let M be an $n \times n$ matrix with entries in $\{0, 1\}$. Use inclusion–exclusion to show that the number of permutations σ of $[n]$ satisfying $M_{i,\sigma(i)} = 1$ for every i is

$$\sum_{J \subseteq [n]} (-1)^{n-|J|} \prod_{i=1}^{n} \left(\sum_{j \in J} M_{ij} \right).$$

Write this same number as a sum over permutations, analogous to the determinant formula but without signs. This sum is called the *permanent* of M. Explain its interpretation as a count of perfect matchings in a bipartite graph.

15. Exactly r properties. With the notation of Problem 13, put $T_j = \sum_{|I|=j} |A_I|$ for $0 \leq j \leq n$. Let N_r count elements of U belonging to exactly r of the sets A_i. For $0 \leq r \leq n$, prove the sieve formula

$$N_r = \sum_{j=r}^{n} (-1)^{j-r} \binom{j}{r} T_j.$$

Hint. First count the contribution of an element belonging to exactly t sets.

16. Bonferroni inequalities. With T_j as in Problem 15, let $U_0 = 0$ and $U_m = \sum_{j=1}^{m} (-1)^{j+1} T_j$ for $1 \leq m \leq n$. Prove that U_m is an upper bound for $|\bigcup_i A_i|$ when m is odd and a lower bound when m is even. Thus the first two truncations give the union bound and the correction obtained by subtracting pairwise intersections.

17. Möbius inversion on subsets. Let f, g be real-valued functions on the subsets of $[n]$, and suppose that

$$f(S) = \sum_{T \subseteq S} g(T) \qquad \text{for every } S \subseteq [n].$$

Prove that

$$g(S) = \sum_{T \subseteq S} (-1)^{|S \setminus T|} f(T).$$

Hint. Substitute the first identity into the second and use Problem 11.

18. Graph colourings. Let $G = (V, E)$ be a finite simple graph and let $q \geq 1$. A proper colouring assigns one of q colours to each vertex so that adjacent vertices have different colours. For $F \subseteq E$, let $c(F)$ be the number of connected components of the spanning graph (V, F), including isolated vertices. Use inclusion–exclusion to prove that the number of proper colourings is

$$P_G(q) = \sum_{F \subseteq E} (-1)^{|F|} q^{c(F)}.$$

Deduce that $P_G(q)$ is a polynomial in q, and check the formula for a triangle.

19. Binary matrices with no empty row or column. For $m, n \geq 1$, count the $m \times n$ matrices with entries in $\{0, 1\}$ such that every row and every column contains a 1. Give a double alternating sum. *Hint.* If specified rows and columns are all zero, how many entries remain free? Check your answer when $m = 1$.

20. Graphs with no isolated vertices. For $n \geq 1$, count the simple undirected graphs on the labeled vertex set $[n]$ that have no isolated vertices. Give a single alternating sum involving binomial coefficients and powers of 2. *Hint.* If a specified set of vertices must be isolated, the remaining vertices may support an arbitrary simple graph. Check the answer for $n = 1, 2, 3$.

21. Use every letter, but never repeat a neighbour. Let $m \geq 1$ and $q \geq 2$. Count the words of length m over a fixed alphabet of q letters such that every letter occurs and no two consecutive letters are equal. Give a single alternating sum. *Hint.* First count words with unequal consecutive letters over a fixed r-letter alphabet, then exclude missing letters. Your formula should also apply when $m < q$.

22. Keep consecutive values apart (H). For $n \geq 2$, count permutations of $[n]$ in which i and $i + 1$ are never adjacent, in either order, for any $i \in [n - 1]$. For $S \subseteq [n - 1]$, let $c(S)$ be the number of maximal nonempty runs of consecutive integers in S, with $c(\varnothing) = 0$. Show that the number of permutations in which i and $i + 1$ are adjacent for every $i \in S$ is

$$2^{c(S)} (n - |S|)!.$$

Use this to obtain an inclusion–exclusion formula. Can you group its terms by $|S|$ and $c(S)$ instead of summing over all subsets?